

FL-03: Context & Knowledge Integration Audit

Author: Rida Eman (CS Senior & ML Intern) | Track: AI Fluency (Build) | Repo: github.com/ridaeman02/flyrank-ml-internship

1. Selected Task & Knowledge Base Overview

Task Focus: Research Paper Summarization & Data Contract Knowledge Base Assembly.

Knowledge Repository: Created tag-structured schema repository at work/context/domain_knowledge.md.

```
<DOMAIN_KNOWLEDGE>
<DATASET_SCHEMA>
- fact_content_daily_performance (report_date, client_hash_id, content_hash_id, gsc_impressions, gsc_clicks,
gsc_avg_position)
- dim_content (content_hash_id, content_created_date, word_count)
- Grain: 9,841,378 daily rows across 70 client domains (March 2026 slice).
</DATASET_SCHEMA>
<PUBLIC_SAFETY_RULES> Pseudonymize all IDs; link data credit to https://flyrank.ai; zero causal claims.
</PUBLIC_SAFETY_RULES>
</DOMAIN_KNOWLEDGE>
```

2. Comparative Context Injection Strategies Audit Table

Context Strategy	Tokens	Fact Accuracy	Hallucination	Public Safety	Verdict
1. No Context	150	20%	80%	High (Generic metrics)	FAIL
2. Raw Dump	4,800	55%	45%	High (Leaked local paths)	FAIL
3. Curated Tags	650	100%	0%	ZERO (Passed)	PASS (Optimal)

3. Strategy Findings & Output Comparison

- **Strategy 1 (No Context):** Model hallucinated standard accuracy numbers (94%) and standard random splits, omitting Precision@50.
- **Strategy 2 (Raw Dump):** Model became overwhelmed by local directory paths and mixed up starter dataset (30k rows) with Hugging Face (9.8M rows).
- **Strategy 3 (Curated Tagged Context):** Model output exact metrics (LogReg Precision@50 = 0.5000 vs Baseline = 0.4200 on client-holdout split) with zero leakage.

4. Reusable Context Management Guidelines

1. **XML Tag Boundaries:** Enclose schemas and safety rules inside explicit XML tags (<DATA_CONTRACT>, <PUBLIC_SAFETY>).
2. **Prune Execution Noise:** Exclude raw local file paths, stack traces, and unorganized git logs before prompt injection.
3. **Anchor Canonical Outlines:** Provide explicit section templates to guarantee structural compliance.